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Tectonics of the Umbria-Marche-Romagna Arc (central northern Apennines, Italy): New paleomagnetic constraints

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Abstract. We present the results of a paleomagnetic study carried out on 32 sites from mainly Messinian clayey sediments distributed throughout the external Umbria-Marche-Romagna Arc (UMRA). These data, together with published results from coeval sediments, demonstrate that this arc is an orocline in its central northern sector. Bending, not well constrained in time, was due to about 15° clockwise rotations of the central part of this arc and to counterclockwise rotations farther north. In this latter area, post-Messinian counterclockwise rotations are of the same amplitude as those calculated for some classic Mesozoic paleomagnetic sections in northern Umbria, suggesting a Plio-Pleistocene age for the rotations reported from the older sequences.

Introduction

The fine-grained Mesozoic pelagic limestones exposed in the Umbria-Marche region (internal UMRA in Figure 1) represent one of the best studied sedimentary sequences in the world, and a wealth of paleomagnetic and magnetostratigraphic data is available for the region. Interpretations of these paleomagnetic data progressively changed, following the improvement in understanding the general tectonic setting of the Apennines and the motions of the nearby plates since the Mesozoic.

The early paleomagnetic data were interpreted assuming that the Umbria-Marche region was essentially autochthonous or "parautochthonous". *Lowrie and Alvarez* [1974, 1975] interpreted their paleomagnetic results in terms of a 30°-40° counterclockwise rotation of the Italian Peninsula, during the Cretaceous, followed by a further 25° rotation after the Eocene. *Klootwijk and VandenBerg* [1975], *VandenBerg and Wonders* [1976], and *VandenBerg et al.* [1978] noted that Cretaceous and Paleogene paleomagnetic directions from the Umbrian sequences fit well with those expected from the African apparent polar wander path (APWP) if a post early Eocene 25° counterclockwise rotation is taken into account. This rotation was extrapolated to the entire Italian peninsula advocating autochthony in northern Umbria (NU in Figure 1).

Channell and Tarling [1975] reported paleomagnetic data from Cretaceous and Paleocene rocks of peninsular Italy and concluded that Africa and a major part of Italy rotated as a whole during Mesozoic and Tertiary times. Only late Cretaceous paleodeclinations from northern Umbria were found to be more westerly with respect to those expected from the African APWP. This discrepancy suggested an additional tectonic deformation of northern Umbria. *Channell et al.* [1978] then provided further paleomagnetic evidence for the allochthonous character of Umbria, stating that data from the Umbria-Marche Arc cannot be extrapolated to the Italian autochthon.

More recently, *Channell et al.* [1992], *Channell* [1992, 1996], and *Van der Voo* [1993] observed that data from Gargano, Istria, the northern Dinaric belt, and the southern Alps do not support any rotation of Adria relative to Africa. Data from northern Umbria were consistent with a late Tertiary 20° counterclockwise rotation of thrusts locally detached from the basement at a décollement in the Triassic evaporites at the base of the sedimentary sequence.

The interpretations concerning the origin of the arcuate shape of the UMRA remain controversial. *Channell et al.* [1978] observed that the more westerly declinations found in the Cretaceous-Eocene "Scaglia" Formation of northern Umbria could be explained with the oroclinal bending of an originally straight fold belt. The same conclusion was proposed by *Eldredge et al.* [1985], who re examined the data of *Channell et al.* [1978]. Subsequently, *Lowrie and Hirt* [1986] and *Hirt and Lowrie* [1988] showed that in contrast to the conclusions of *Eldredge et al.* [1985], there is no constant relationship between fold axis orientation and paleomagnetic declination. Moreover, *Lowrie and Hirt* [1986] studied the early Cretaceous "Maiolica" limestones in the same region and showed that the nonuniformity of paleodeclinations throughout the arc was not related to the orientations of the fold axes. They concluded that the Umbria-Marche Arc was not, strictly speaking, an orocline.

The geodynamic interpretation of the paleomagnetic data from the UMRA is problematic due to (1) the difficulty in relating Mesozoic paleomagnetic data to the Neogene tectonic deformation, (2) the poor definition of the African APWP, and (3) the structural position of most of the sites, located in the severely folded and faulted internal UMRA.

Other than paleomagnetic data on Mesozoic and early Tertiary pelagic sections, only two studies dealt with

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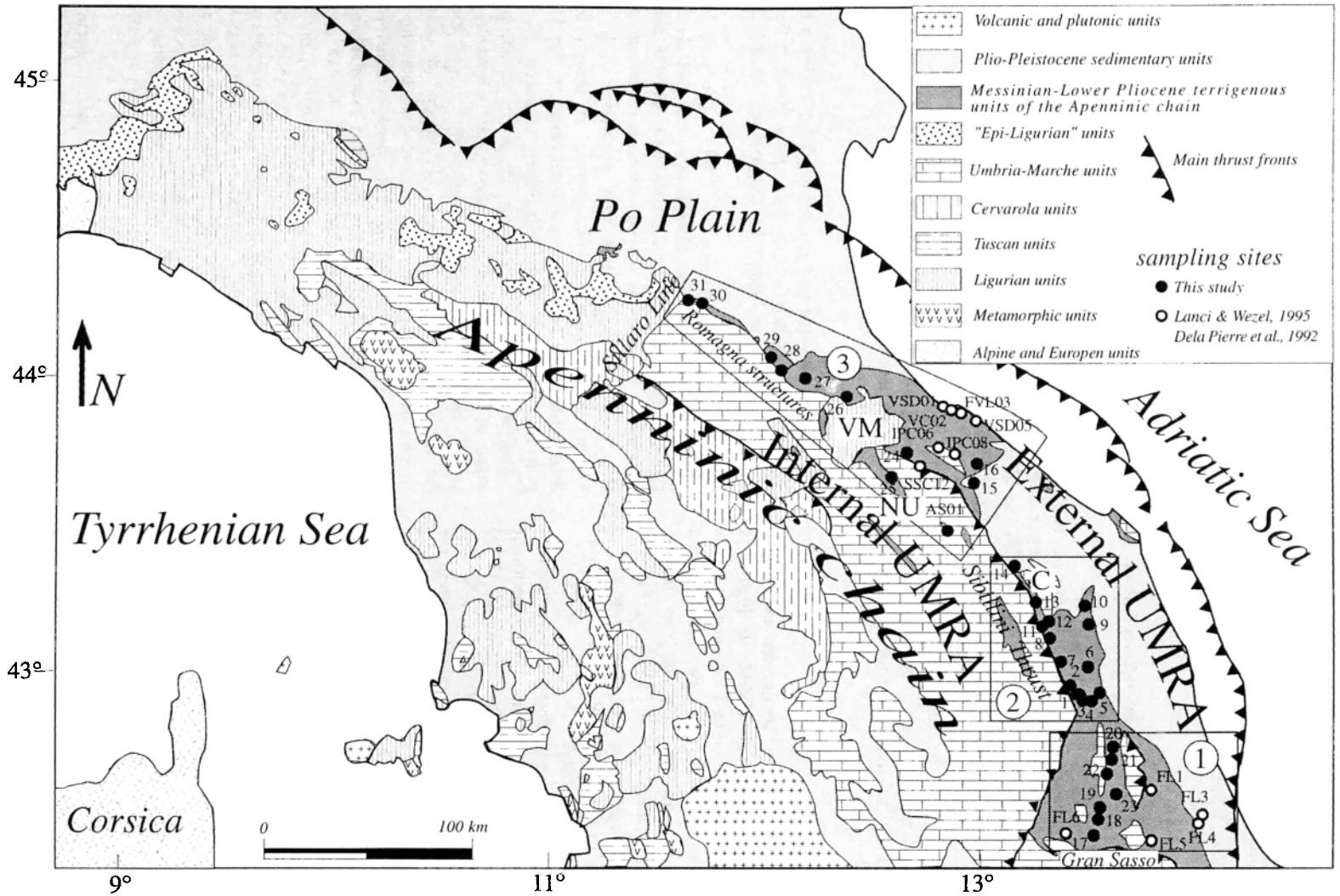


Figure 1. Schematic map of the central northern Apennines and locations of sampling sites. Boxed areas are 1, Acquasanta domain; 2, Sibillini-Cingoli domain; 3, Marche-Romagna domain. Abbreviations are UMRA, Umbria-Marche-Romagna Arc; VM, Val Marecchia unit; NU, northern Umbria; C, Cingoli anticline. Only the sequential numbers are reported for the sampling sites from this study (except for site AS01); they are referred in the text with the prefix "MR".

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paleomagnetism of Messinian-Pleistocene sediments from external UMRA (Figure 1). *Dela Pierre et al.* [1992] did not observe rotations in the Messinian-early Pliocene formations from the southern edge of the arc (Acquasanta domain, Figure 1). On the other hand, *Lanci and Wezel* [1995] showed that Messinian sediments from the northern Marche underwent 10° to 90° of anticlockwise rotations.

In this paper we present new paleomagnetic results from upper Miocene sediments cropping out in the external, less deformed, UMRA, which give direct information about the Plio-Pleistocene tectonic history. The results of this study provide new kinematic constraints for the reconstruction of the northern Apennines system.

Geological and Tectonic Setting

The northern Apennines consists of several paleogeographic domains, arranged in arcuate thrust sheets with an eastward (Adriatic) vergence [*Consiglio Nazionale delle Ricerche*, 1989]. The regional direction of the main structures gradually changes from N-S in the southern part of the belt to NW-SE or WNW-ESE in the northern part (Figure 1). Extension of the Tyrrhenian margin was synchronous with thrust emplacement along the external Apenninic chain. Both extensional and compressional fronts migrated since middle-late Miocene up to Pleistocene toward the Adriatic [*Elter et al.*, 1975]. Extensional tectonics dissected the already formed Apenninic chain, producing several extensional basins mainly NW-SE, infilled by "neoautochthonous" marine and continental sequences. In Tuscany, widespread extension accompanied thinning of the crust, high heat flow, and uprise of magmatic bodies. The active extensional structures, with a well-documented historical and recent seismicity, are mostly located in the internal UMRA, along the edges of intermontane basins.

The UMRA represents the external domain of the chain (Figure 1). It consists of Meso-Cenozoic sedimentary units incorporated in several thrust sheets stacked during the Messinian and Plio-Pleistocene [i.e., *Pieri and Groppi*, 1981; *Consiglio Nazionale delle Ricerche*, 1989; *Castellarin*, 1992; *Castellarin et al.*, 1992; *Calamita et al.*, 1994]. Mesozoic-early Tertiary pelagic limestones and marls are mainly exposed in the internal UMRA, whereas Messinian to Pleistocene terrigenous deposits characterize the external UMRA. Toward the north, UMRA units are tectonically overlain by the Ligurian allochthonous terrains along the Sillaro line [*Castellarin et al.*, 1992]. The Ligurian units also tectonically cover the UMRA at the southern edge of the folds and thrusts of the Romagna (Val Marecchia unit, Figure 1). Toward the south, the external UMRA disappears below the Gran Sasso thrust front [*Ghisetti and Vezzani*, 1991]. From the Acquasanta domain to the Cingoli anticline, the internal UMRA is separated from the external UMRA by the Sibillini thrust front. North of the Cingoli anticline, the internal UMRA is separated from the external UMRA by a simple flexure and by minor thrust faults (Figure 1).

In the external UMRA, the Meso-Cenozoic sequence is covered by Messinian to Pleistocene terrigenous sediments and crops out only along a few structures. In the central and southern sector of the external UMRA, the Messinian sediments are mainly represented by the Laga Formation, a foredeep flysch sequence formed by repeated alternations of arenaceous turbidites and pelagic clays and marls. The turbidites probably formed after erosion and fast accumulation of Alpine detritus in the Messinian foredeep of the advancing Apennines tectonic wedge [*Ricci Lucchi*, 1982]. The Laga Formation reaches a thickness of about 3000 m in the Acquasanta domain and thins out toward the north, disappearing north of the Cingoli anticline (Figure 1). Gypsarenite beds are present in this formation, and they are conventionally used as a distinctive marker to individuate three (pre evaporitic, evaporitic, and post evaporitic) members. In the northern external UMRA, the Messinian sediments consist of euxinic clays around the Tortonian-Messinian boundary, followed by the evaporitic deposits of the "Gessoso-Solfifera" Formation and by the "Argille a Colombacci" Formation. The Argille a Colombacci Formation (mentioned in the following as the Colombacci Formation) is composed of clays interlayered with rare beds of chemical limestones and is etheropic with the post evaporitic member of the Laga Formation. This formation is also present in internal UMRA, along narrow synclines elongated parallel to the main thrust fronts and deposited in Messinian piggy-back basins [*Patacca et al.*, 1990]. Finally, in the external UMRA, Plio-Pleistocene marine sediments crop out along the Adriatic coast, generally unconformably covering the older sequences.

Sampling

We collected 308 cylindrical core samples, 25 mm in diameter, from 32 sites (outcrops), all (except sites AS01 and MR25) along the external sector of the UMRA (Figure 1). Cores were drilled using an ASC 280E petrol-powered portable drill and oriented in situ with a magnetic compass. The samples were collected from fine-grained blue-grey marine clayey strata. All but three of the sites are early to late Messinian in age; among the others, two (MR30, MR31) are Tortonian-Messinian, and one (MR14) is early Pliocene. We collected 8 to 11 oriented cores, spaced in different beds. Bedding was well defined at each site.

The southernmost seven sites (MR17 to MR23) were sampled in the pre evaporitic member of the Laga Formation, early Messinian in age, except site MR21, which belongs to the middle Messinian evaporitic member. These sites are located along the N-S structures of the Acquasanta-M. dei Fiori thrusts. In the same area, *Dela Pierre et al.* [1992] reported results from five sites in the Laga Formation (FL1, FL2, FL5 to FL7) and two (FL3, FL4) in the clays of the "Cellino" Formation (early Pliocene). All these sites will be mentioned in the following as Acquasanta sites.

The Laga Formation was sampled at 13 sites (MR01-MR13), from the structures at the front of the Sibillini thrust, up to the Cingoli anticline (Figure 1). Close to the Cingoli Anticline, one additional site (MR14) was sampled in early Pliocene clays transgressive on the Messinian sequence. Sites MR07, MR11, and MR12 are located in the pre evaporitic member, sites MR01-MR05 and MR13 are in the evaporitic member, and sites MR06 and MR08-MR10 are in the post evaporitic member. All are between thrust sheets and folds striking NNW-SSE. They will be identified in the following as Sibillini-Cingoli sites.

Finally, 11 sites were sampled in the northern sector of the UMRA, mainly in the clayey levels of the late Messinian Colombacci Formation, except two (MR30, MR31) sampled in the euxinic clays at the Tortonian-Messinian (T/M)

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boundary and another (MR29), sampled in early Messinian sequences (Figure 1). Five sites (MR15, MR16, MR24, MR25, AS01) are just south of the Val Marecchia unit. Two of these (MR25, AS01) are in the internal UMRA, close to the classical Mesozoic limestone sections paleomagnetically studied during the past 20 years. The other six sites (MR26-MR31) are in the Romagna structures, between the Val Marecchia unit and the Sillaro line. South of the Val Marecchia unit, in external UMRA, *Lanci and Wezel* [1995] also studied 12 sites in the Colombacci Formation (Figure 1). All of these, which are located in WNW-ESE striking structures, will be referred in the following as Marche-Romagna sites.

Laboratory Procedures

Core samples were cut into one to four specimens, 22 mm high. All magnetic remanence measurements and demagnetization experiments were carried out in the magnetically shielded room of the paleomagnetic laboratory at the Istituto Nazionale di Geofisica (Rome, Italy). Measurements of remanence directions and intensities were performed using a JR-5 spinner magnetometer.

Rock Magnetism

Magnetic mineralogy analyses were perfomed on at least one specimen for each site. We characterize the magnetic carriers using several sources of information (Figure 2): (1) isothermal remanent magnetization (IRM) acquisition curves, using a pulse magnetizer with a maximum field of 1.6 T; (2) removal of the IRM produced at 1.6 T by progressive application of a increasing back-field and estimation of the coercivity of remanence; (3) thermal demagnetization, using a magnetically shielded electrical oven, of a composite IRM produced according to the method of *Lowrie* [1990]; (4) alternating field demagnetization of the IRM at 1.6 T, using a Molspin tumbler AF demagnetizer; (5) hysteresis loops, using a Molspin vibrating sample magnetometer; and (6) thermomagnetic curves expressing the variation of the low-field magnetic susceptibility with temperature, using a CS-2 furnace apparatus connected to a KLY-2 susceptibility bridge.

Except for six sites, four from the Colombacci Formation and the two sites from the euxinic clays at the T/M boundary, hysteresis loops and thermomagnetic curves generally only indicate a paramagnetic behavior and were not useful to the identification of the remanence carriers. The magnetic mineralogy was thus mostly identified by experiments on artificially imparted remanences.

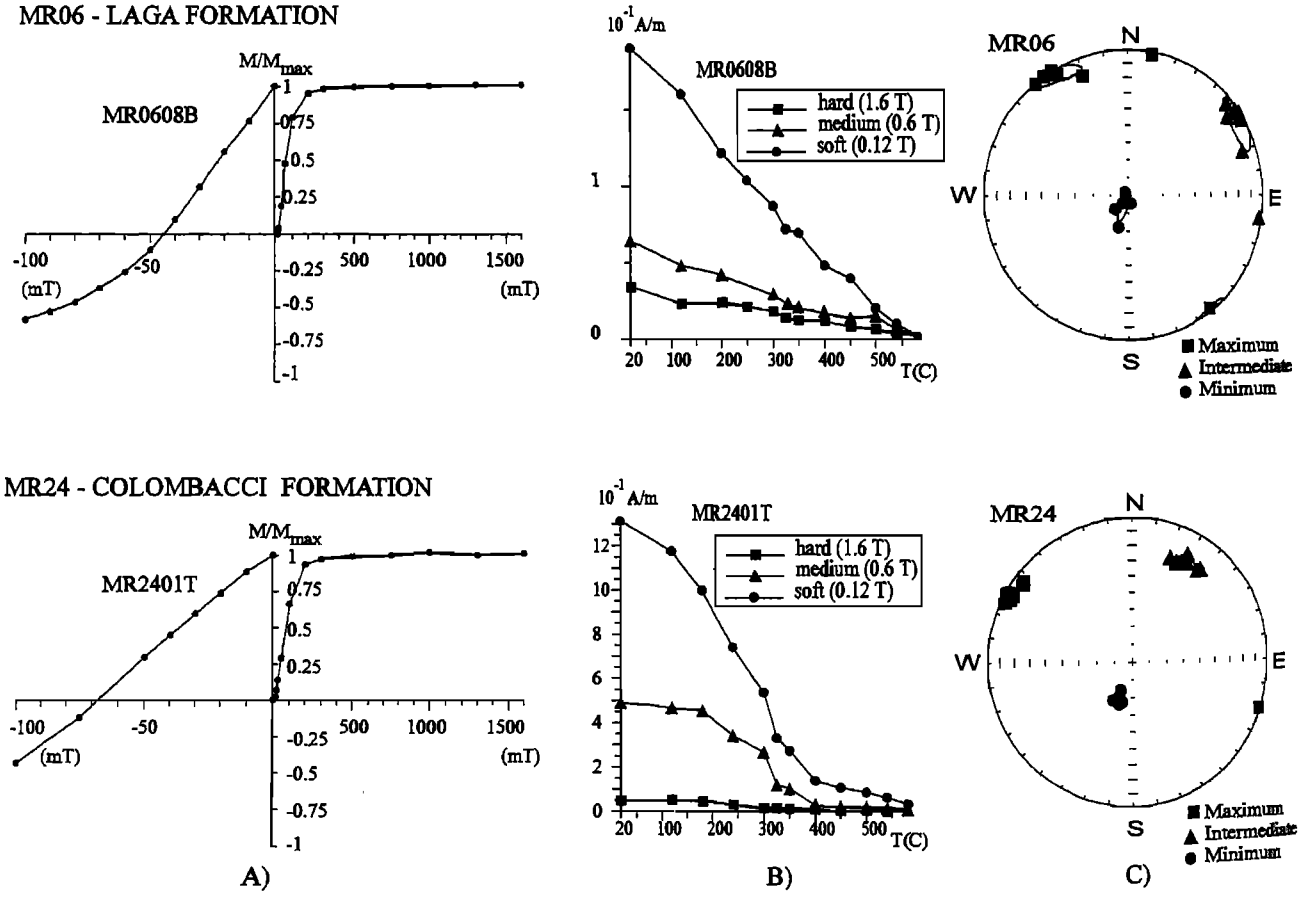


Figure 2. Results of magnetic mineralogy (Figures 2a and 2b) and AMS analyses (Figure 2c) for two representative sites from the Laga and Colombacci formations. (a) (right) Acquisition of normalized IRM and (left) back field demagnetization curve. (b) Thermal demagnetization of a three-component IRM. (c) Schmidt equal-area projection, lower hemisphere-geographic coordinates; the ellipses indicate the 95% confidence regions around the principal susceptibility axes.

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The IRM acquisition and demagnetizations showed that the remanence is held by low-coercivity magnetic minerals (Figure 2a). The thermal demagnetization of a composite IRM indicate that at some sites two magnetic fractions can be distinguished in the soft-coercivity curve, with different maximum unblocking temperatures, in the range 300-350°C and around 580°C, respectively (Figure 2b). The whole set of data indicates that the main magnetic carriers are magnetite and ferrimagnetic iron sulphides. Magnetite has been identified at all sites. The presence of magnetic iron sulphides is evident at 17 sites, both from the Laga Formation, the Colombacci Formation, the euxinic clays at the T/M boundary, and the early Messinian clays. At two sites (MR30 and AS01), an indeterminate soft-coercivity magnetic phase with a sharp maximum unblocking temperature in the range 350-400°C was observed.

The anisotropy of the low-field magnetic susceptibility (AMS; measured using the KLY-2 bridge) also was studied for all the specimens (Figure 2c). Only the main results that are relevant to this paleomagnetic study will be discussed here. The AMS is well resolved at each site and indicates that the magnetic fabrics of the studied formations are typical of weakly deformed (clay-rich) sediments, in which the plane of magnetic foliation is parallel to bedding and the maximum susceptibility axis is parallel to the (local) fold axis. Details about the AMS analyses will be discussed elsewhere.

Demagnetization of the Natural Remanent Magnetization and Vector Analysis

For each site, two pilot specimens from the same core were progressively demagnetized, one by thermal heating, the other by alternating magnetic fields. The variation of the low-field magnetic susceptibility was monitored throughout the thermal treatment to resolve the onset of possible changes in the original magnetic mineralogy. Demagnetization data were plotted both on orthogonal demagnetization diagrams [*Zijderveld*, 1967] and equal-area projections and evaluated using the standard principal component analyses [*Kirschvink*, 1980]. Pilot results were largely comparable, but the thermal treatment was preferred for the demagnetization of all the remaining specimens.

Samples from four sites (MR09 and MR11 from the Laga Formation, MR29 from early Messinian clays, and MR14 from early Pliocene clays) were found to be weakly magnetized (natural remanent magnetization NRM less than $1x10^{-5}$ A/m) and were excluded from further laboratory treatments. All specimens from the other sites, in which the room temperature NRM ranges between $1x10^{-2}$ and $5x10^{-5}$ A/m, were thermally demagnetized in 5 to 15 steps (8-10 in average). Thermal treatment was stopped when NRM decreased to about one tenth of the initial value or when random changes of the paleomagnetic direction appeared during heating (generally, after 300-400°C).

For about 10% of the studied samples (distributed across almost all the sites and mainly from MR17, MR18, and MR19), after the elimination of the viscous component at 200°C, two components of magnetization coexist until incoherent changes of paleomagnetic directions appear at 300-400°C (Figure 3a). In this case the best fit great circle to the remanence directions above 200°C was calculated.

In the remaining 90% of the studied samples, a characteristic remanent magnetization (ChRM) could be isolated (and calculated using at least three demagnetization steps) after removal of the viscous component at about 200°C (Figures 3b-3d). For most of the sites (16 from the Laga Formation and 3 from the Colombacci Formation), maximum unblocking temperatures rarely exceed 400°C (Figure 3b). In two sites from the Colombacci Formation (MR25, MR26), a ChRM can be isolated only up to 380°C, when NRM and low-field magnetic susceptibility increases indicate formation of new ferromagnetic phases (Figure 3c). At this temperature, about 70% of the room temperature NRM was removed by the thermal treatment. Finally, at seven sites, three from the Colombacci Formation (MR24, MR27, MR28), two from the beds at the T/M boundary (MR30, MR31), and two from the Laga Formation (MR05, MR06), a complete demagnetization

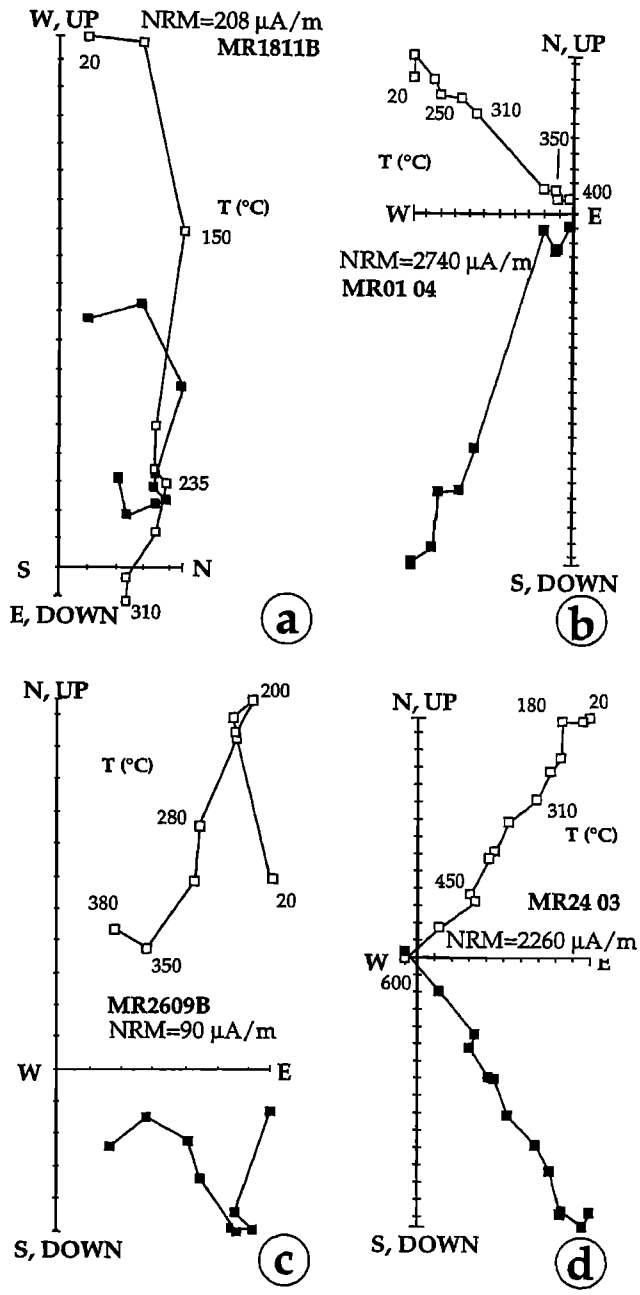


Figure 3. Vector diagrams of typical thermal demagnetization data, tilt corrected. Open and solid symbols represent projections on the vertical and horizontal planes, respectively.

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Table 1. Paleomagnetic Directions From External Umbria-Marche-Romagna Arc

Site	Form.	SG,deg	SA,deg	$n_1,n_2/N$	Db,deg	Ib,deg	Da,deg	Ia,deg	k	α_{95},deg	Ref.
MR01	L2	336	334	9,0/10	125.9	-45.9	196.0	-23.9	43	8.0	1
MR02	L2	336	333	6,1/10	105.0	-41.8	206.9	-35.8	38	10.0	1
MR03	L2	336	338	6,1/11	121.2	-47.2	206.0	-32.8	27	11.9	1
MR04	L2	345	341	5,1/10	200.3	-19.7	186.3	-19.2	21	15.2	1
MR05	L2	338	327	6,2/10	133.8	-39.6	183.4	-43.2	21	12.5	1
MR06	L3	338	328	8,0/11	170.1	-33.2	173.6	-23.1	55	7.5	1
MR07	L1	331	323	3,2/09	132.7	-81.3	191.5	-67.1	96	8.4	1
MR08*	L3	339	19	3,2/09	206.9	-34.7	205.8	-26.9	16	21.0	1
MR10	L3	330	323	6,3/09	183.4	55.7	199.7	-42.8	47	7.7	1
MR12	L1	339	329	2,2/09	16.8	23.1	354.5	46.8	61	13.7	1
MR13*	L2	321	318	4,1/10	137.2	-38.0	171.3	-29.4	11	24.6	1
MR15	CO	297	292	10,0/10	122.1	-66.3	145.0	-61.6	100	4.8	1
MR16	CO	297	278	9,0/09	130.5	-47.2	153.5	-56.9	82	5.7	1
MR17*	L1	359	355	1,8/09	156.7	-80.4	149.5	-69.5	264	3.5	1
MR18*	L1	2	353	4,7/11	155.4	-68.1	225.5	-71.5	137	4.1	1
MR19*	L1	2	350	2,7/09	146.3	-75.8	245.9	-75.7	87	6.0	1
MR20	L1	342	327	8,0/09	6.9	22.7	358.6	21.9	95	5.7	1
MR21	L2	351	335	8,1/10	153.5	-41.7	188.1	-43.2	31	9.4	1
MR22	L1	2	7	8,0/08	187.2	-55.7	187.7	-49.7	246	3.5	1
MR23	L1	348	330	9,0/09	197.8	-60.9	162.0	-51.3	154	4.2	1
MR24	CO	297	298	10,0/10	137.5	-49.6	154.1	-36.2	109	4.7	1
MR25	CO	308	305	10,0/10	108.7	-49.7	173.7	-48.4	36	8.2	1
MR26	CO	285	293	9,0/10	105.8	-52.4	128.1	-71.5	61	6.6	1
MR27	CO	285	287	9,0/09	109.8	-30.7	109.8	-30.7	85	5.6	1
MR28	CO	317	307	9,0/09	171.9	-45.4	172.6	-35.4	92	5.4	1
MR30	TM	289	293	10,0/10	352.3	65.9	339.0	43.6	52	6.7	1
MR31	TM	289	308	8,2/10	145.1	-67.3	159.1	-47.0	34	8.4	1
AS01	CO	310	315	5,1/09	167.2	-56.9	162.4	-47.6	66	8.5	1
FL1	L1	360	/	11,0/11	131.7	-20.5	183	-48	/	9.6	2
FL3	CE	348	/	11,0/11	335.5	48.3	351	45	/	15.2	2
FL4	CE	348	/	11,0/11	348.5	56.7	13	41	/	6.7	2
FL5	L2	350	/	10,0/10	130.9	-32.9	176	-42	/	9.0	2
FL6	L1	10	/	13,0/13	354.9	49.3	360	27	/	5.9	2
VSD01	CO	300	/	3,0/15	181.0	-8.7	166.9	-36.1	519	5.4	3
VC02	CO	300	/	12,0/12	192.2	-3.6	166.5	-60.5	358	2.3	3
FVL03	CO	300	/	5,0/05	201.9	-44.8	183.8	-54.7	973	2.5	3
VSD05	CO	300	/	9,0/09	166.1	-36.8	148	-55.5	193	3.7	3
PC06	CO	300	/	7,0/07	121.1	-67.2	145.4	-51	70	7.3	3
IPC08	CO	305	/	9,0/09	169.9	-53.1	103.9	-72.4	74	6.0	3
SSC12	CO	305	/	10,0/10	182.7	-57.4	167.7	-55.7	56	6.5	3

Formations (Form.), TM Tortonian-Messinian boundary beds; L1, L2, L3 preevaporitic, evaporitic, and postevaporitic members of the Laga Flysch early, middle, and late Messinian in age respectively; CO Colombacci Formation (late Messinian); CE Cellino Formation (early Pliocene). S structural (fold axis) direction at a site calculated from geological maps (SG), and from AMS data (SA) (see text). $n_1,n_2/N$ number of stable directions, number of great circles/total number of studied samples at a site. D, I site mean declinations and inclinations calculated before (Db, Ib) and after (Da, Ia) tectonic correction. k and α_{95}, statistical parameters after *Fisher* [1953]. References (Ref.), (1) this paper; (2) *Dela Pierre et al.* [1992]; and (3) *Lanci and Wezel* [1995].
*Discarded sites (see text).

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occurs at 550-600°C (Figure 3d). In about half of these samples, a strong decay of the NRM intensity is observed between 300 and 400°C. At the sites where both magnetite and iron sulphides were identified with magnetic mineralogy analyses, the paleomagnetic directions do not change over all the demagnetization, showing that the two magnetic phases carry the same stable component.

Site-mean paleomagnetic directions were calculated using *Fisher*'s [1953] statistics in the 13 sites where only stable ChRM were isolated or using the *McFadden and McElhinny* [1988] method in the 15 sites where both ChRMs and remagnetization circles were fitted to the demagnetization paths. Almost all the mean directions are well defined, the α_{95} value being generally lower than 10° (Table 1). Two sites (MR08 and MR13, Table 1) having large within-site directional scatter (α_{95}>20°) were discarded from regional means and from the following geodynamical considerations.

Results

Twenty-six site-mean paleomagnetic directions were obtained by the procedures discussed above. Additional results are reported by *Dela Pierre et al.* [1992] for the Acquasanta domain, three directions from the Laga Formation and two for the early Pliocene Cellino Formation (Table 1, two sites are scattered). The precision parameter k was not reported by the authors. Moreover, 7 out of 12 well-constrained site-mean directions obtained by *Lanci and Wezel* [1995] from the Colombacci Formation just to the southwest of the Val Marecchia unit are also reported on Table 1. A total of 38 reliable sites are thus available, with ages ranging from the Tortonian-Messinian boundary to early Pliocene.

All the site-mean paleomagnetic directions from external UMRA are plotted in Figure 4. After bedding correction, only five sites have a normal polarity, one from the clays at the T/M boundary (MR30), two from the Pre-Evaporitic Laga Formation (MR20, FL6), and the two early Pliocene sites (FL3, FL4). This may suggest that the sedimentation of the postevaporitic member of the Laga Formation and the Colombacci Formation occurred entirely during the 3r late Messinian reverse polarity chron [*Cande and Kent*, 1992].

A relatively large scatter of the data is observed, both before and after bed dip correction. After bedding correction, the site mean distribution is Fisherian but a large scatter appears for both declinations and inclinations (Figure 4). The possible sedimentary and tectonic origins of this scatter will now be discussed, considering the reasons for inclinations and declinations dispersions.

Since no Cenozoic APWP is available for the Adriatic promontory, the observed paleomagnetic directions cannot be compared to those expected for the Adriatic foreland. However, most authors [*Channell et al.*, 1992; *Channell*, 1992, 1996; *Van der Voo*, 1993] considered that Adria followed the African movements during Meso-Cenozoic times and that Africa is not believed to have significantly rotated or greatly latitudinally drifted during the Neogene [*Besse and Courtillot*, 1991]. For these reasons, as reference direction we adopted the geocentric axial dipole (GAD) field direction (in both normal and reverse polarity state), assuming a negligible difference between Messinian and Present times.

Analysis of Inclinations

Tilt-corrected site-mean paleoinclinations are shown in Figure 5. In most of the sites, inclinations are significantly shallower than the reference; maximum inclination flattening can be observed in the Laga Formation (generally 20° and up to 40° of shallowing), with no significant difference between the three members of the formation. On the other hand, most of the sites from the Colombacci Formation have an inclination error lower than 10°, even if four sites show about 30° difference from the reference value. Significant inclination errors (of about 20°) can be also observed in the sites from Cellino Formation (early Pliocene) and from clays at the T/M boundary. The inclination shallowing observed in the Acquasanta sites of the Laga Formation and Cellino

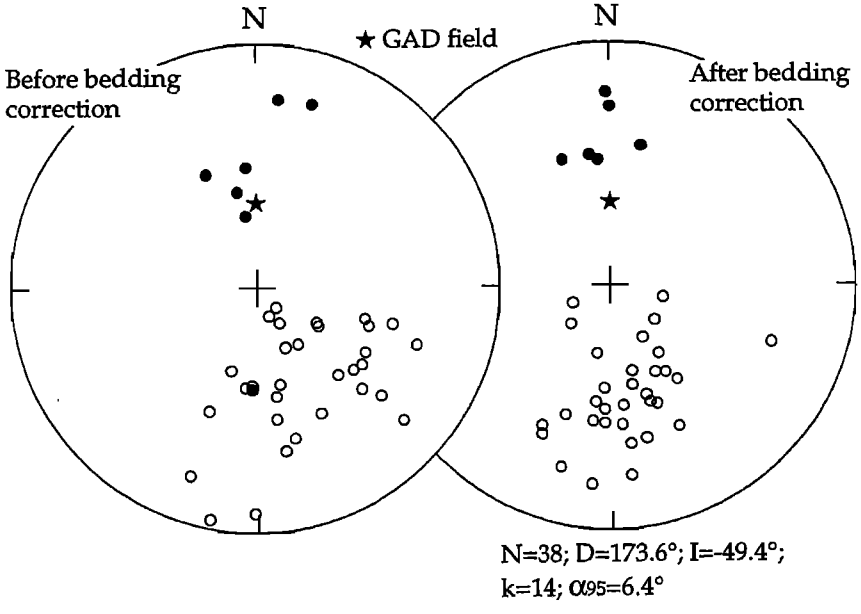


Figure 4. Equal-area projection of the site mean directions from the external UMRA. Open (solid) circles represent projection onto upper (lower) hemisphere. GAD, geocentric axial dipole (normal polarity).

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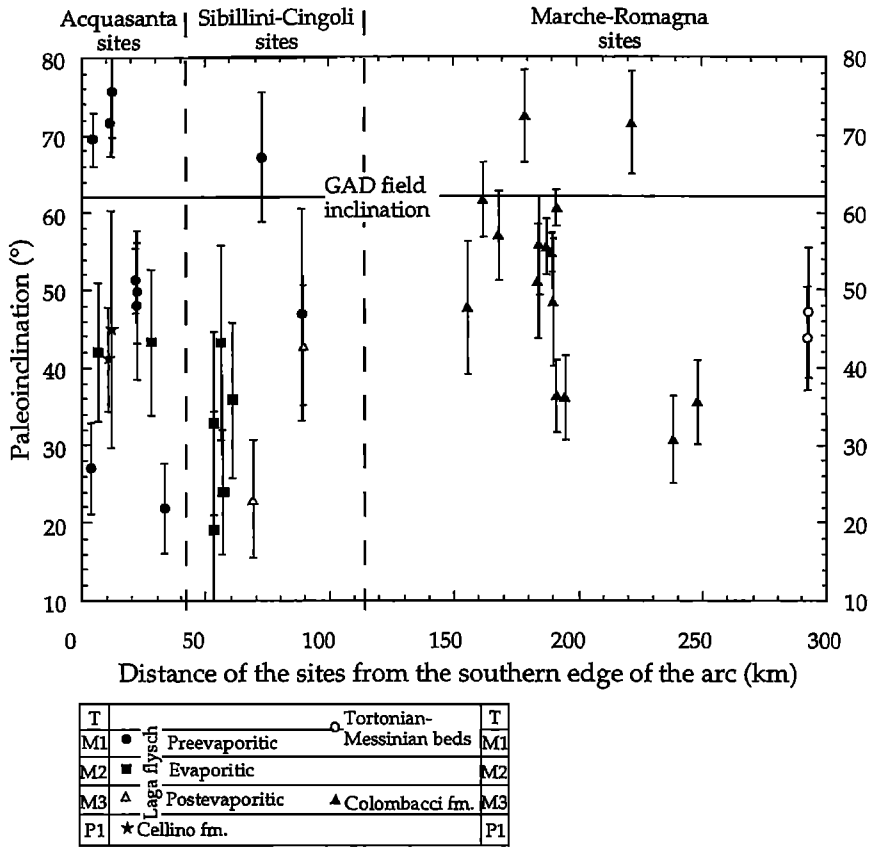


Figure 5. Paleomagnetic site-mean inclinations along the external UMRA. Distances are computed from the Gran Sasso thrust front. Error bar half length for inclination is α_{95}. Ages are T, Tortonian; M1, M2, and M3: early, middle, and late Messinian, respectively; P1, early Pliocene.

Formation was interpreted by *Dela Pierre et al.* [1992] as related only to depositional effects.

Six site-mean directions are steeper than expected (Figure 5). Bedding was always well resolved, so that this feature does not arise from erroneous bedding corrections. Remagnetization is similarly unlikely for the three sites (MR07, MR26, and IPC08) in the Sibillini-Cingoli and Marche-Romagna domains, since in situ site-mean directions are far from the GAD direction. For the three sites in the Acquasanta domain (MR17-MR19), the majority of the samples did not yield a stable ChRM during thermal demagnetization and, thus site-mean directions have been calculated mostly using the convergence point of several remagnetization circles (Table 1). The paleomagnetic mean directions from these three sites are better grouped before (D=152.8°; I=-74.8°; k=162) than after (D=203.6°; I=-76.5°; k=30) bedding correction. For all these reasons they have been considered anomalous and have been discarded for further analyses.

We think that the large inclination shallowing reflects mostly compaction-related effects in clayey formations [*Arason and Levi*, 1990; *Deamer and Kodama*, 1990]. The larger inclination errors found in the Laga Formation, which reached about 3000 m of total thickness, could reflect the great amount of compaction experienced by the clayey layers. In spite of this evidence, relations between burial compaction and inclination shallowing are not straightforward. As a matter of fact, about the same inclination errors can be observed in the three members of the Laga Formation, which are located at different depths of the stratigraphic sequence.

Flattening of the sediments is also indicated by the AMS data. At all the sites, a well-defined magnetic foliation parallel to the bedding planes predominates the magnetic fabric, with the parameter F (Kint/Kmin) between 1.012 and 1.113 (1.06 in average). However, no clear relation is found between inclination shallowing and the AMS parameter F for the Laga Formation and the euxinic clays at the T/M boundary (Figure 6). A linear relation is found at the Colombacci Formation sites. This is likely due to a greater contribution of the ferromagnetic minerals to the AMS at some sites in the Colombacci Formation. In consideration of such a complex pattern, it seems clear that tilt-corrected inclinations from external UMRA are not useful to constrain the paleolatitude of northern Apennnines during Messinian times.

Analysis of Declinations

Tilt corrected paleomagnetic declinations along the external UMRA are shown in Figure 7. With very few exceptions, a coherent pattern of declinations can be observed in the same sectors of the arc.

The Acquasanta sites have paleodeclinations generally oriented around N-S. The sites north of the Acquasanta domain show a rather gradual change in the declinations. Eastward declinations of about 10°-20° are documented in the Sibillini-Cingoli domain (Figure 7). On the contrary, north of the Cingoli Anticline only westward declinations were found. Generally, the sites south of the Val Marecchia unit have westward declination of about 20°. The two sites just north of the Val Marecchia unit (MR26, MR27) have larger (~60°)

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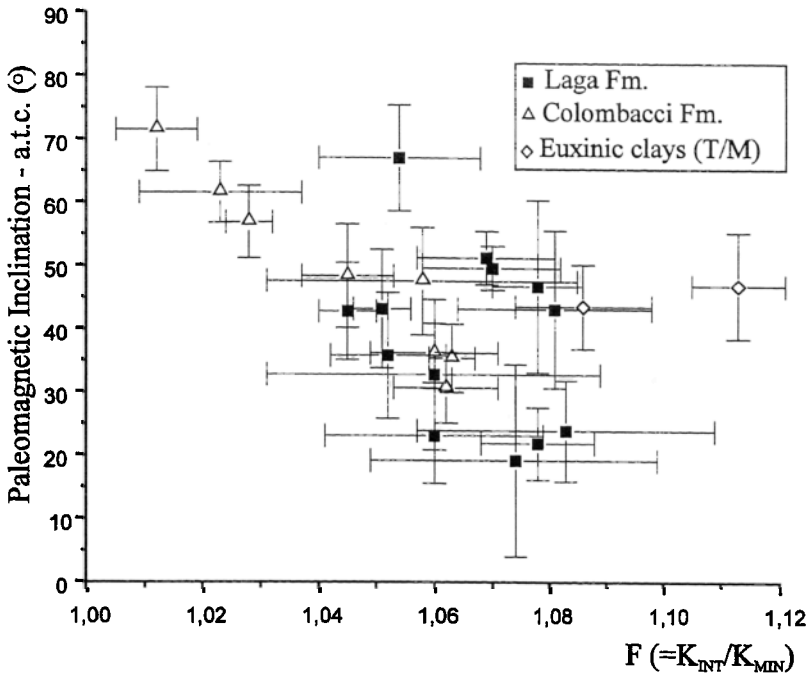


Figure 6. Relation between site mean paleomagnetic inclination (atc, after tectonic correction) and site mean magnetic foliation F. Error bars for inclination are the α_{95} values. Error bars for magnetic foliation are the standard deviations of the site-mean values.

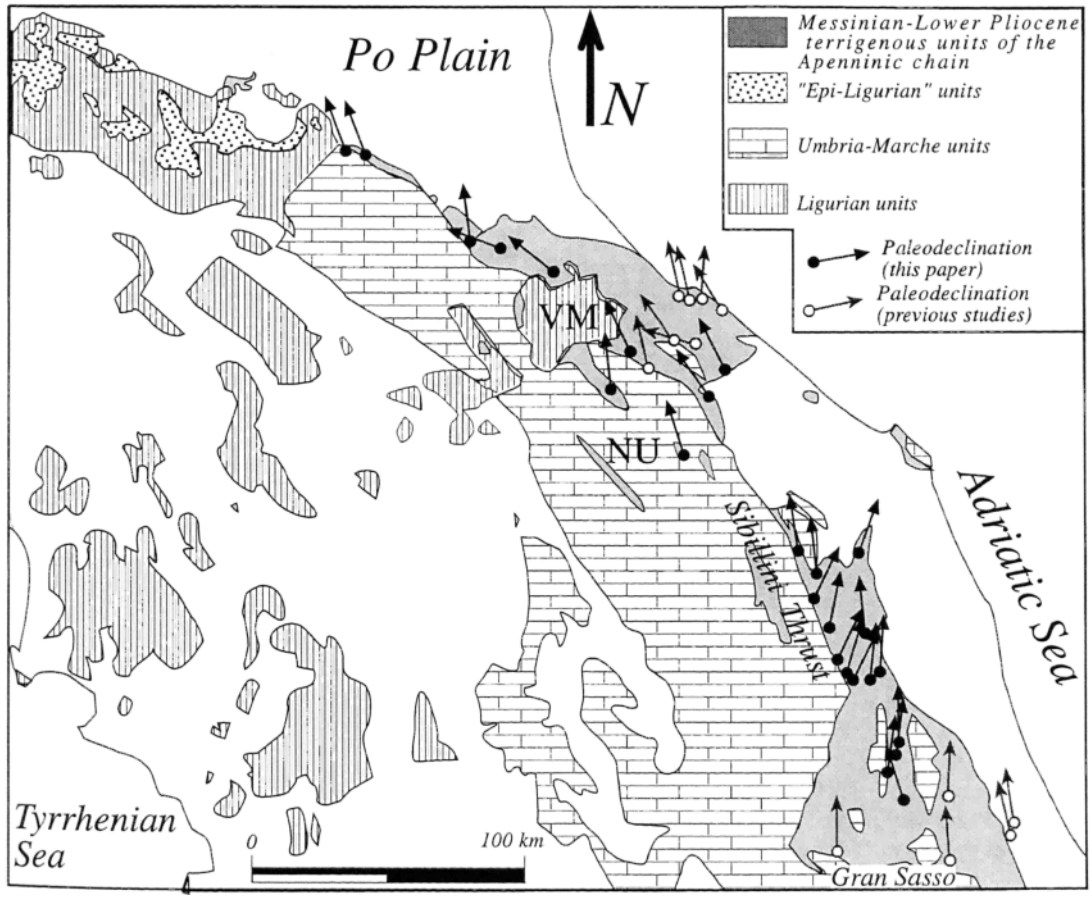


Figure 7. Tilt-corrected paleomagnetic declinations (all transferred to normal polarity) along the external UMRA. Abbreviations are VM, Val Marecchia unit; NU, northern Umbria.

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westward declinations, while westward declinations of about 20° were again observed farther north up to the Sillaro line.

The relation between paleomagnetic declinations and structural directions were investigated using the method originally proposed by *Schwartz and Van der Voo* [1983] for the Appalachians and successively applied to the internal UMRA by *Eldredge et al.* [1985], *Lowrie and Hirt* [1986], and *Hirt and Lowrie* [1988]. In Figure 8, the declination deviations are plotted against fold axis deviations, as determined from geological maps (Figure 8a and Table 1) and from AMS data (Figure 8b and Table 1). In these diagrams, we chose 360° as the reference declination (Dr), so that positive (negative) declination deviations indicate clockwise (counterclockwise) rotations. We used 315° as the reference fold axis direction, which is a perfect NW-SE trend, since directions spanned in the external UMRA range from roughly N-S (Acquasanta domain) to roughly E-W (just north of the Val Marecchia unit). The reference values for declinations and fold axis directions are not critical, the important point being that they must be kept constant.

Figure 8a shows that all of the data from external UMRA, except for those coming from the not rotated Acquasanta domain, are well correlated (correlation coefficient R=0.74). The slope of the regression line calculated (excluding the Acquasanta sites) is significantly different from zero slope at the 95% confidence level following the statistical *t* test. The *t* test of the slope (m=1.0) of the regression line (excluding the Acquasanta sites) compared to zero slope gives t= 5.34. This is greater than the critical *t* value at the 99% significance level (for number of data N=26), $t_{99\%}$= 2.49. Figure 8b shows that the correlation improves (N=19; R=0.83; t= 6.20; $t_{99\%}$=2.57) when the fold axis directions are estimated by the magnetic lineation trends. Indeed, we believe that the use of the in situ site-mean direction of the principal maximum susceptibility axis (K_{max}) provides the best estimation of the local fold axis direction (relying on the observed parallelism between K_{max} and structural trends in the external UMRA). This method is more precise than deriving structural directions from large scale geological maps and it overcomes the problem of estimating the structural direction at sites where bedding is weakly tilted and/or in structures with plunging structural axis.

In both Figures 8a and 8b, the slope of the best fit line approaches the value of 1, that is the expected value for an orocline which was originally perfectly straight. The calculated structural direction of the belt (according to Figure 8b) before it has undergone any bending is 323°. The Acquasanta structures strike roughly N-S and have not undergone rotation.

Regional Mean Paleomagnetic Directions and Field Tests

Sites from the Acquasanta domain yield a regional mean paleomagnetic direction defined by N=9, D=0.1°, I=41.4°, k=44.6, and α_{95}=7.8° (Figure 9). Paleomagnetic directions from normal and reverse polarity sites are virtually antipodal after bedding correction (positive reversal test of classification Rc according to *McFadden and McElhinny* [1990]. The data pass the *McFadden* [1990] fold test at the 99% confidence level ($\xi_{\text{in situ}}$= 5.72; ξ_{unfolded}= 0.494; $\xi_{99\%}$= 4.85; minimum ξ was observed at 93% of complete unfolding), suggesting both effective laboratory magnetic cleaning and the primary nature of the characteristic remanent

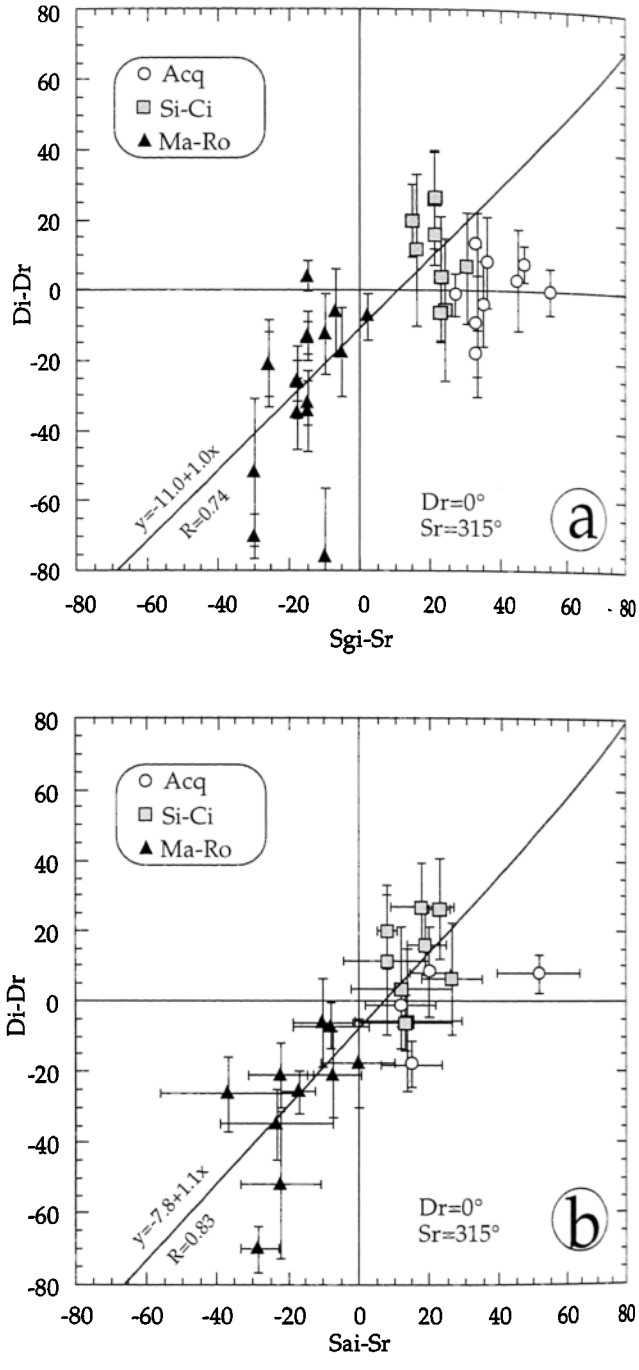


Figure 8. Declination deviations relative to fold axis deviations. Di is the observed declination at a site (transferred to normal polarity), Dr is the reference declination. The fold axis at a site is determined from geological maps [*Servizio Geologico d'Italia*, 1968, 1969a, b; *Centamore*, 1986; *Centamore et al.*, 1992] or from AMS data and termed Sgi and Sai in Figures 8a and 8b, respectively. Sr is the reference fold axis direction. Error bars for declination are the α_{95}/cosI site values. Error bars for fold axes (in Figure 8b) are the e_{12}/cosI_{AMS} site values. e_{12} is the semiangle of the 95% confidence ellipse around principal maximum susceptibility axis in the magnetic foliation plane; I_{AMS} is in situ inclination of the principal maximum susceptibility axis at a site. Acq, Acquasanta sites; Si-Ci, Sibillini-Cingoli sites; Ma-Ro, Marche-Romagna sites. The best fit lines are calculated by linear regression analysis to the data from the Sibillini-Cingoli and Marche-Romagna sites.

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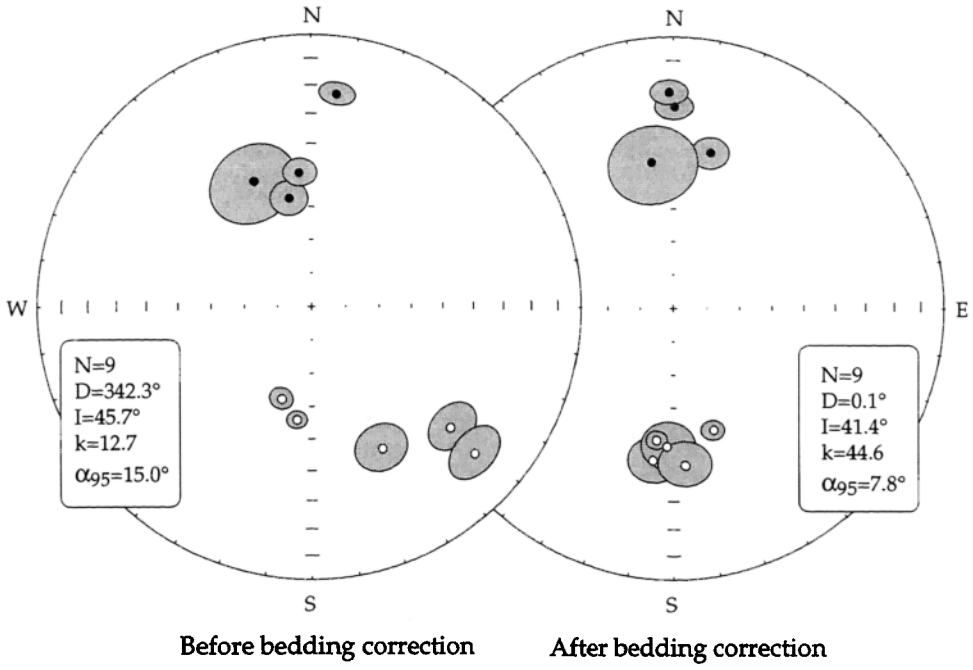


Figure 9. Equal-area projection of the site mean directions from the Acquasanta domain. Open (solid) circles represent projection onto upper (lower) hemisphere. Shaded areas are the projections of the α_{95} cone about the mean direction.

magnetization. These data show that in the southern sector of the external UMRA, the major Acquasanta and Montagna dei Fiori thrusts and the minor neighboring structures formed during early Pliocene as N-S ridges and did not successively rotate. This concurs with the findings of *Dela Pierre et al.* [1992].

Mean paleomagnetic directions from the Sibillini-Cingoli and Marche-Romagna domains are shown in Figure 10. Several lines of evidence prove that these directions do not result from recent remagnetization and have a geodynamical significance:

1. In situ directions are all (but the one from site MR30) far from the normal polarity GAD field direction and the data distribution is non-Fisherian (according to the M_u/M_e test [see *Fisher et al.*, 1987; *Tauxe et al.*, 1991].

2. Even if only 2 out of the 26 site-mean directions show normal polarity, they are almost antipodal to the reverse ones after tilt correction (positive reversal test of the type Rc).

3. A bootstrap fold test [*Tauxe and Watson*, 1994] indicates that data are most tightly grouped at about 83% unfolding, excluding both 0 and 100% unfolding as the appropriate coordinate systems at the 95% level of confidence. This is likely a consequence of oroclinal bending, as explained below.

Figure 10 shows that a large scatter of declinations exists even after a simple bedding correction (the data are, however, Fisher distributed when all are transferred to the reverse hemisphere). Clockwise rotated sites are concentrated in the Sibillini-Cingoli domain, whereas 20° to 60° counterclockwise rotations were found in sites from Marche-Romagna area. Finally, we applied the "orocline" correction to the tilt-corrected paleomagnetic site mean directions. This consists in rotating the site-mean declinations by an angle equal to the difference between the local fold axis direction in a site and a reference direction in order to align the local fold axes to a regional trend [*Schwartz and Van der Voo*, 1983; *Hirt and Lowrie*, 1988]. After this kind of correction, the grouping of paleomagnetic directions from oroclinal belts is expected to improve. The plot of corrected directions from central northern external UMRA with respect to the 323° reference structural direction (which is the nonrotated direction calculated in Figure 8b) is also shown in Figure 10. The scatter of the data decreases, and this further supports the conclusion that the central and northern sectors of external UMRA derived their present-day geometry from an oroclinal bending mechanism.

Discussion

Paleomagnetic evidence for Neogene and Quaternary rotations of crustal units of different senses and variable amount has already been observed in other sectors of the central and southern Apennines [*Alfa et al.*, 1988; *Sagnotti*, 1992; *Dela Pierre et al.*, 1992; *Scheepers et al.*, 1993; *Scheepers*, 1994; *Mattei et al.*, 1995]. We now show that local Plio-Pleistocene rotations also affected the structures of the UMRA and determined their present-day geometry.

The mean post-Messinian counterclockwise rotations of about 20° measured in the Marche-Romagna area, south of the Val Marecchia unit (Figure 7 and Table 1), are of the same amplitude as those calculated for some close northern Umbria Mesozoic sections, recently reevaluated by *Channell* [1992, 1996] and *Van der Voo* [1993], and support the hypothesis of a Plio-Pleistocene age for the rotations measured in the older sequences. However, local variations of paleodeclinations were also reported for Mesozoic to Early Tertiary sequences in northern Umbria adjacent sections [*Channell*, 1992], confirming that the detail of the overall rotational pattern is likely to be affected by local deformation of individual

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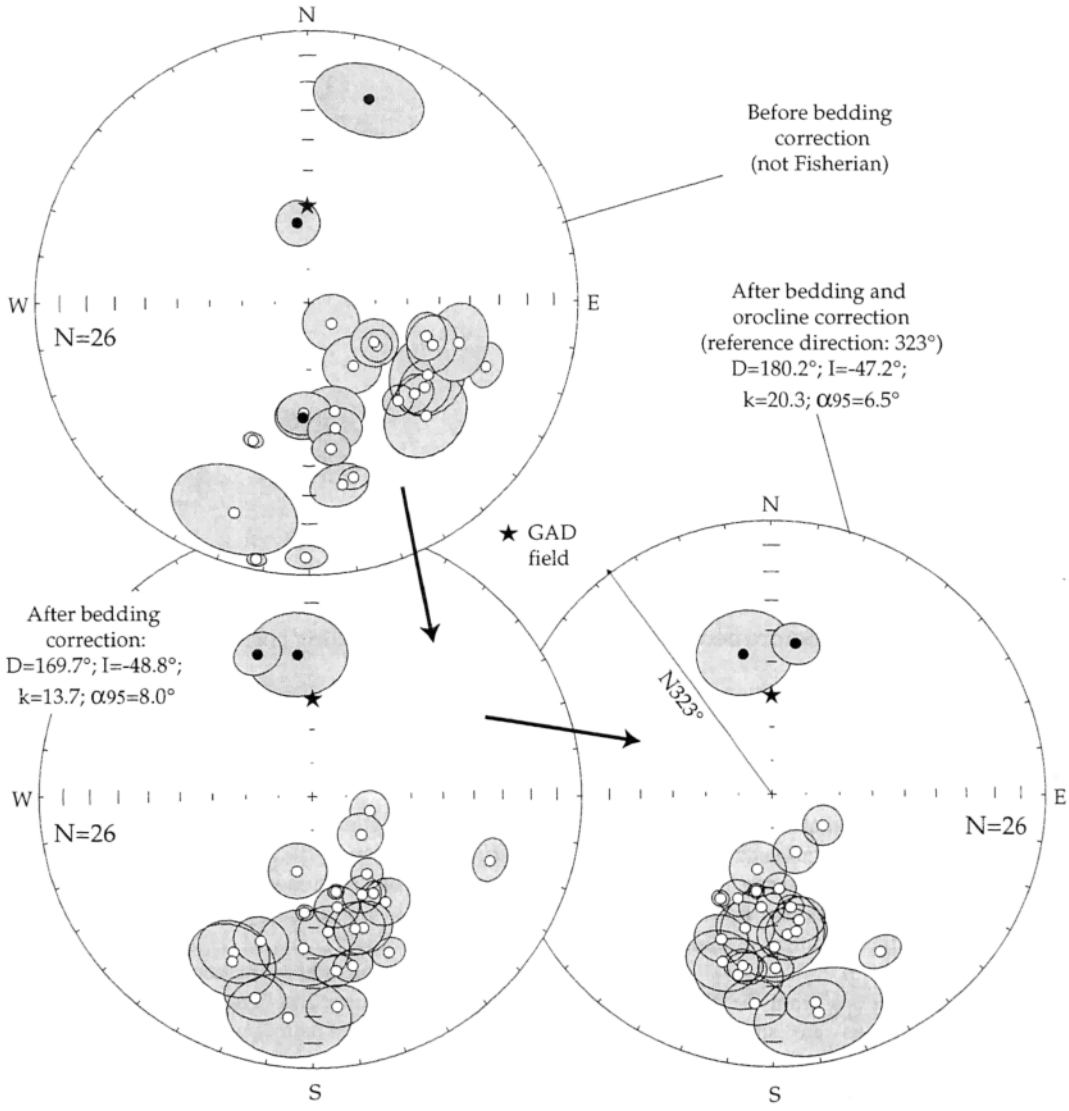


Figure 10. Equal-area projection of the site mean directions from the Sibillini-Cingoli and Marche-Romagna domains. Open (solid) circles represent projection onto upper (lower) hemisphere. Shaded areas are the projections of the α_{95} cone about the site-mean directions.

structures. Extrapolations to large regions and to different time intervals are extremely questionable.

From the Sibillini thrust up to the Sillaro line, the UMRA is an orocline. This conclusion is different from that drawn by *Hirt and Lowrie* [1988] after a paleomagnetic study of early Cretaceous limestones from internal UMRA. The internal UMRA is structurally much more complicated than the external UMRA due to the presence of Plio-Pleistocene normal and strike-slip faults which postdate the emplacement of folds and thrusts. Moreover, concerning the choice of the local structural directions in the paleomagnetic sites, *Hirt and Lowrie* [1988] used both the bedding strikes and the fold axes extrapolated by field measurements, whereas we suggest that the site-mean magnetic lineations from AMS analyses of fine-grained sediments provide a more precise tool for evaluation of fold axis directions.

Conclusions

New paleomagnetic results demonstrate that the central northern Umbria-Marche-Romagna Arc is an orocline. The arcuate shape of the belt formed either from bending of a late Miocene-early Pliocene straight belt oriented roughly 320° or progressively with the thrust emplacement. The central part (Sibillini-Cingoli domain) of the arc rotated about 15° clockwise, whereas the northern part (Marche-Romagna domain) rotated counterclockwise. Counterclockwise rotations shift from 20° to 60° from south to north of the Val Marecchia unit, returning to about 20° farther northward. In the southern sector of the arc (Acquasanta domain), a N-S sunken sector drove the Messinian flysch into the foredeeps at the front of the Latium-Abruzzi nappes, which were also 320° in strike [*Mattei et al.*, 1995] and shifted westward with respect to the northern Apennines.

Extensional basins are located just behind the UMRA (Figure 1) and infilled by late Miocene-Pleistocene sediments [*Ambrosetti et al.*, 1978], which did not subsequently rotate [*Mattei et al.*, 1996]. It is not possible to establish whether extensional tectonics in the Tuscan Tyrrhenian margin, eventually linked to the collapse of the overthickened Apennines wedge, forced the previously rectilinear belt to bend, or if bending produced back arc extension. Anyhow,

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bending and extension behind the arc are both genetically needed.

Timing of the post early Pliocene bending of the arc (presumably contemporaneous to rotations in the central Apennines) is not well constrained but is probably associated with major nappe displacements in the external front of the belt.

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